Beyond Answer Generation: Scaffolding Metacognitive Thinking via LLM-Driven Socratic Dialogue

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***Abstract*— While Large Language Models have rapidly become powerful providers of knowledge in education, developing higher-order thinking skills such as metacognition remains among the most difficult educational challenges. Unlike knowledge acquisition, these skills require learners to actively monitor, evaluate, and regulate their own reasoning processes. This study presents a conversational learning framework that operationalizes Socratic-style inquiry through LLM-driven dialogue to cultivate learners into better thinkers by strengthening metacognitive and Computational Thinking practices. Instead of providing answers, the system analyzes learners' explanations derived from PISA science tasks and generates adaptive reflective questions that guide reasoning refinement. Deployment with 20 university students across three weekly sessions showed increased thinking proficiency, reflected by reduced reliance on metacognitive scaffolding. The average number of system-generated reflective prompts required for task completion decreased from 19.45 to 16.35 turns across repeated system use, indicating improved independent reasoning during explanation construction. These findings suggest that LLMs can be repositioned from knowledge providers toward inquiry partners that support the development of thinking skills in AI-mediated learning environments.**

Keywords*— *Large Language Models, Metacognition, Computational Thinking, Socratic Dialogue, AI-Supported Learning

I. INTRODUCTION

The emergence of Large Language Models (LLMs) has rapidly transformed educational technology by enabling conversational systems capable of providing explanations, feedback, and problem-solving assistance through natural language interaction. Recent studies show that LLM-based systems can function as intelligent tutoring agents supporting learners through adaptive dialogue and contextualized explanations [1]. As generative AI becomes increasingly integrated into learning environments, students can obtain immediate solutions with minimal cognitive effort, reshaping learning behaviors and knowledge acquisition processes [2]. Although such accessibility improves learning efficiency and scalability, concerns have emerged that heavy reliance on AI-generated answers may reduce active cognitive engagement, allowing learners to bypass reasoning processes essential for self-regulated learning and metacognitive development [3]. Prior research in artificial intelligence in education further suggests that excessive automation of problem solving may shift learners from active knowledge constructors toward passive consumers of generated solutions, potentially weakening reflective thinking during learning activities [4].

While LLMs demonstrate strong capability in providing explanations and domain knowledge, access to information alone does not guarantee the development of higher-order cognitive skills. Educational research distinguishes knowledge acquisition from cognitive skill formation, emphasizing that effective learning requires learners to actively regulate, evaluate, and refine their own thinking processes [5], [6]. Metacognitive abilities support critical thinking by enabling learners to assess reasoning quality, justify decisions, and adapt understanding across unfamiliar contexts [7]. The growing integration of generative AI introduces the possibility of cognitive offloading, where learners delegate cognitive and regulatory processes to external technological systems rather than engaging in internal reasoning [8]. Although such delegation may improve efficiency, excessive reliance on automated assistance can reduce reflective engagement during problem solving. Recent evidence suggests that sustained dependence on generative AI may lead to metacognitive laziness, where learners rely on AI systems to perform monitoring and evaluative functions typically associated with self-regulated learning [9]. Consequently, learning environments focused primarily on answer generation risk limiting opportunities for students to actively engage in reasoning processes necessary for developing transferable thinking skills.

One pedagogical approach for promoting reflective thinking and metacognitive engagement is dialogic learning through guided questioning, commonly associated with the Socratic method. Rather than transmitting knowledge directly, Socratic instruction encourages learners to articulate reasoning, examine assumptions, and refine understanding through iterative questioning. Research shows that structured questioning can stimulate deeper cognitive processing by prompting learners to justify responses, evaluate alternatives, and monitor the coherence of their reasoning [10]. Such dialogic interaction functions as cognitive scaffolding, supporting learners in actively constructing understanding rather than passively receiving information [11]. Questioning-based instruction therefore shifts learning from answer acquisition toward reasoning development and metacognitive regulation. Studies in

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learning sciences further indicate that reflective questioning promotes self-explanation and critical evaluation, both strongly associated with improved problem-solving performance and conceptual understanding [12].

Motivated by these principles, this study proposes an alternative paradigm for employing LLMs in education by positioning the model not as an answer generator but as a facilitator of metacognitive dialogue structured around Computational Thinking (CT) processes. The proposed system engages learners in conversational interaction centered on open-ended problem-solving tasks where students construct responses rather than request solutions. Instead of evaluating answers solely by correctness, the LLM analyzes student responses to identify weaknesses in CT dimensions such as problem decomposition, reasoning structure, abstraction, and logical organization. When limitations are detected, the system generates adaptive follow-up questions that prompt learners to reconsider their reasoning strategies and reflect on how solutions are constructed. Through this questioning-driven interaction, the LLM operationalizes Socratic scaffolding within a scalable conversational environment, using Computational Thinking as a structured framework for eliciting metacognitive regulation. By shifting AI interaction from solution delivery toward reflective engagement in problem-solving processes, the proposed framework aims to mitigate cognitive offloading and reduce metacognitive laziness associated with answer-centric AI usage, thereby sustaining learners' active participation in reasoning and solution construction.

II. Related Work

A. Large Language Models for Learning Support

Recent studies have increasingly explored the use of Large Language Models (LLMs) as learning support tools capable of functioning as conversational tutors and instructional assistants. Kasneci et al. examined the educational opportunities and challenges of deploying LLMs such as ChatGPT, highlighting their ability to provide adaptive explanations and personalized academic assistance through natural language interaction while cautioning against potential learner overreliance on automated responses [1]. Similarly, Zhai investigated the role of generative AI in next-generation learning environments and reported that LLM-based interaction enables learners to obtain immediate explanatory support, fundamentally reshaping learner–technology interaction patterns [2].

Beyond conceptual discussions, empirical and design-oriented studies have explored practical implementations of LLM-supported tutoring systems. Holmes et al. discussed how AI-driven educational technologies support scalable instructional assistance by providing automated feedback and continuous learning support outside traditional classroom settings [13]. Studies examining conversational AI deployment further report improvements in learner engagement and task efficiency when LLMs are used as instructional assistants. However, existing implementations predominantly employ LLMs to generate explanations or solutions, positioning the model primarily as a knowledge provider rather than a facilitator of reasoning processes [14], [15]. Collectively, these studies demonstrate the effectiveness of LLMs in enhancing accessibility to learning support and instructional feedback. Nevertheless, the dominant interaction paradigm remains answer-oriented, leaving open questions regarding how LLM-based systems may actively sustain learners' cognitive engagement and reflective reasoning during problem-solving activities.

B. Technology-Supported Metacognitive Learning

Technology-enhanced learning research has long investigated mechanisms for supporting learners' metacognitive regulation during learning activities. Metacognition, defined as the monitoring and control of one's cognitive processes, plays a central role in effective learning and adaptive problem solving [5], [6]. Consequently, intelligent learning environments increasingly incorporate scaffolding strategies that encourage learners to plan, monitor, and evaluate their understanding throughout learning tasks.

Azevedo and colleagues demonstrated metacognitive scaffolding through the MetaTutor system, which used adaptive pedagogical agents to provide metacognitive prompts during science learning. Their results showed that learners receiving such scaffolding exhibited stronger self-regulated learning behaviors and deeper conceptual understanding compared with those receiving content support alone [16]. Similarly, Roll et al. found that intelligent tutoring systems offering metacognitive feedback improved students' help-seeking strategies and reflective problem-solving, highlighting the value of process-oriented guidance rather than correctness feedback [17]. W. Na Chai et al. further showed that flexible metacognitive scaffolding in computer-mediated inquiry environments enhanced learners' awareness of planning and evaluation during problem solving [18]. Related work has also introduced tools that capture learners' thinking through structured writing, enabling systems to externalize reasoning processes and support reflective monitoring during learning [19].

Despite these advances, AI-supported learning environments introduce new challenges for metacognitive engagement. Cognitive offloading refers to delegating cognitive processes to external technologies, potentially reducing learners' internal monitoring effort [8]. In generative AI contexts, Fan et al. further suggest that reliance on such systems may lead to metacognitive laziness, where learners depend on AI to perform evaluative and monitoring functions typically associated with self-regulated learning [9]. These concerns highlight the need to design AI-supported environments that sustain learners' active participation in metacognitive regulation.

III. Methodology

This study proposes a conversational learning framework designed to support students' metacognitive development through guided interaction with a Large Language Model (LLM). The system operationalizes metacognitive training by structuring learner–AI dialogue around Computational Thinking (CT) processes during problem-solving activities. Rather than functioning as an answer-generating tutor, the LLM assumes the role of a metacognitive facilitator that encourages learners to examine and refine their reasoning. Learning tasks are retrieved according to learners' domains of interest and presented as open-ended problems requiring explanation-based responses. Student answers are analyzed with respect

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to reasoning structure, after which the system generates reflective questions intended to stimulate reconsideration of problem-solving strategies. Through iterative dialogue, learners are encouraged to monitor, evaluate, and revise their own thinking processes. This interaction forms a continuous metacognitive induction loop in which reasoning refinement emerges from guided questioning rather than corrective feedback as shown in Figure 1.

A. Learning Interaction Design

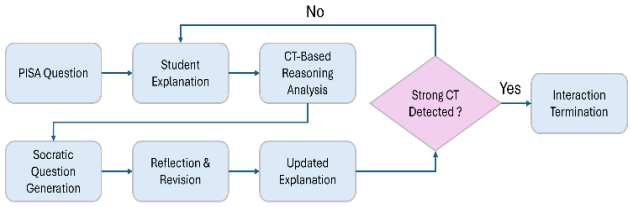


Fig 1. The proposed metacognitive learning process driven by CT-aligned Socratic questioning

The learning interaction is designed to promote metacognitive development by engaging learners in reflective dialogue centered on their own problem-solving explanations. In this study, metacognition refers to learners' ability to monitor, evaluate, and regulate their reasoning processes while constructing solutions. Such abilities develop when learners actively articulate their thinking, examine underlying assumptions, and revise explanations through guided reflection rather than receiving direct answers. Accordingly, the proposed interaction emphasizes reasoning construction and self-evaluation instead of correctness verification.

Metacognitive acquisition within the proposed framework occurs through iterative cycles of explanation and reflection. Learners are required to provide explanation-based responses to open-ended problems retrieved from the learning repository. By externalizing reasoning through written explanation, learners make their thinking processes observable and available for analysis. The conversational agent facilitates reflection by posing questions grounded in the learner's own explanation, encouraging reconsideration of how conclusions are formed. To preserve learner responsibility for reasoning evaluation, correctness feedback and direct solutions are intentionally withheld, preventing cognitive offloading and sustaining active cognitive engagement. A key feature of the interaction design is the use of Computational Thinking (CT) as a framework for guiding metacognitive dialogue. Learners' explanations are analyzed across core CT dimensions, including decomposition, pattern recognition, abstraction, algorithmic reasoning, and evaluation. Based on this analysis, the conversational agent dynamically selects questioning strategies targeting specific limitations in learners' reasoning processes. Rather than correcting responses, the agent functions as an inquiry facilitator that elicits computational thinking practices and promotes awareness of reasoning strategies.

Table 1 presents the CT-aligned adaptive questioning framework used in the system, mapping reasoning conditions in learner explanations to corresponding metacognitive intentions and questioning roles. For example, oversimplified explanations trigger prompts encouraging problem decomposition, while limited abstraction prompts questions promoting conceptual generalization. When reasoning conflicts with available evidence or reflects conceptual misunderstanding, questioning strategies aim to expose inconsistencies and prompt reconsideration without providing direct corrective answers. Through repeated exposure to explanation-contingent questioning strategies, learners gradually internalize computational thinking schemes and become increasingly capable of evaluating and refining their own reasoning processes. The interaction proceeds iteratively as learners revise explanations in response to reflective prompts, forming a continuous explanation–reflection cycle that supports metacognitive regulation.

The level of instructional support is adaptively regulated throughout interaction. After each learner revision, the system re-examines demonstrated computational thinking characteristics to determine whether reflective scaffolding remains necessary. When learners exhibit coherent abstraction, structured decomposition, and logically consistent reasoning, questioning frequency is progressively reduced. The interaction concludes once sufficient evidence

Table I. Computational Thinking–Aligned Metacognitive Questioning Strategies

Core CT Dimension	**Reasoning Condition Observed**	**Typical Issue in Explanation**	**Questioning Role**	**Example Reflective Prompt**
Decomposition	Oversimplified explanation	Treats problem as single entity	Component Separation	"Can you describe the different parts involved in this situation?"
Pattern Recognition	Surface reasoning	Focuses on keywords instead of relationships	Pattern Identification	"What similarities or relationships do you observe here?"
Abstraction	Case-bound reasoning	Cannot generalize principle	Abstraction Induction	"What general rule explains this outcome?"
Algorithmic Thinking	Illogical process	Steps contradict mechanism	Process Verification	"Does this step logically follow from the previous one?"
Evaluation / Debugging	Completely incorrect explanation	Conclusion conflicts with evidence	Conflict Induction	"Which information in the passage supports your conclusion?"
Evaluation / Debugging	Correct but shallow	Limited justification	Elaboration Prompt	"Why does this factor determine the result?"
Generalization	Partial understanding	Correct answer without transfer	Generalization Prompt	"Would this reasoning work in a different situation?"
Metacognitive Regulation	Premature closure	Stops reasoning early	Reflection Extension	"Is there another possible explanation?"

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of structured reasoning is observed, reflecting the pedagogical principle of scaffolding fading in which instructional support is withdrawn as learner competence emerges. Consequently, learning occurs through sustained participation in reflective reasoning activities supported by adaptive conversational inquiry rather than answer provision.

B. LLM Configuration and Knowledge Retrieval

The proposed conversational framework employs a Large Language Model (LLM) to operationalize the metacognitive questioning strategies described in Section 3.1. Rather than functioning as a general-purpose question answering system, the LLM is configured to assume the role of a metacognitive facilitator responsible for analyzing learners' explanations and generating reflective questions aligned with Computational Thinking processes. To achieve this role specialization, the system integrates controlled prompting strategies together with a retrieval-augmented content mechanism that constrains task presentation and conversational context.

1) LLM Role Configuration

The conversational agent in the proposed framework is implemented using Qwen3-7B, an open-weight Large Language Model selected for its strong multilingual reasoning capability and efficient deployment in educational environments. The model operates locally within the system to enable controlled interaction and consistent behavioral configuration during learning sessions. Unlike conventional educational chatbots designed for question answering, the LLM is explicitly configured to function as a metacognitive facilitator. The model performs three primary functions: (1) interpretation of learner explanations, (2) identification of computational thinking characteristics demonstrated in responses, and (3) generation of adaptive reflective questions aligned with the questioning strategies summarized in Table 1. To ensure consistent role behavior, instruction-based prompting is employed to constrain the model toward inquiry-oriented interaction while preventing direct disclosure of correct answers.

Prompt: Metacognitive Facilitator Role Instruction

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You are an educational conversational agent whose role is
to support students' thinking processes.

Your objective is NOT to provide answers or judge
correctness. Instead, analyze the student's explanation
and identify limitations in their reasoning using
Computational Thinking principles, including:
- problem decomposition
- abstraction
- logical reasoning
- step-by-step process understanding
- evaluation of reasoning
If weaknesses are detected, ask a guiding question that
helps the student reconsider or improve their explanation.

Rules:
1. Do not reveal the correct answer.
2. Do not say whether the answer is correct or incorrect.
3. Ask questions related to the student's explanation.
4. Encourage reflection and clarification.
5. Stop questioning when the student demonstrates clear
and well-structured reasoning.
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The system prompt (Metacognitive Facilitator Role Instruction) defines the conversational role of the LLM as an inquiry agent responsible for stimulating learners' reasoning processes rather than providing instructional explanations. The prompt explicitly instructs the model to analyze learners' explanations through Computational Thinking dimensions and to generate questions that guide learners toward reflection and self-correction.

During interaction, learner responses together with task descriptions retrieved via the RAG module are supplied as contextual input to the model. The LLM evaluates explanation structure and generates concise reflective prompts encouraging clarification, justification, abstraction, or verification of reasoning. Generation parameters are configured to prioritize stable analytical dialogue, ensuring that responses remain focused on questioning rather than explanation delivery. Through this role-oriented configuration, Qwen3-7B operates as an adaptive questioning agent capable of inducing metacognitive regulation while maintaining alignment with the pedagogical interaction design described in Section 3.1.

2) Retrieval-Augmented Knowledge Source

Learning tasks used in the proposed system are maintained within a Retrieval-Augmented Generation (RAG) framework to ensure pedagogical consistency during conversational interaction. Instructional materials are derived from publicly released Programme for International Student Assessment (PISA) science assessment items, which are designed to evaluate reasoning and problem-solving competencies in authentic real-world contexts. The adoption of standardized assessment materials ensures that learning activities reflect established evaluation practices rather than artificially constructed instructional questions.

To support interaction within a text-based conversational environment, only PISA items consisting entirely of textual information were included, while tasks requiring graphical interpretation, visual diagrams, or image-based scenarios were excluded. Each selected PISA content unit may contain multiple assessment questions and is stored in a semi-structured format consisting of three components: content, question, and scoring hint. The content component provides contextual information describing the scientific scenario, while the questions represent reasoning tasks derived from the same scenario. When multiple questions exist within a content unit, the system randomly selects one during task initialization to reduce repetition effects and encourage engagement with reasoning processes rather than memorization. For PISA items originally presented in multiple-choice format, answer options are intentionally omitted and the question is reformulated with an explanatory requirement (e.g., "and why?") to elicit justification-based responses. Each task also includes a scoring hint, representing expert-informed reasoning expectations specific to the question. Rather than serving as answer keys, these hints guide the LLM in diagnosing potential limitations in learners' explanations and generating reflective questions aligned with the CT-adaptive questioning framework summarized in Table 1.

Scoring hints are applied strictly on a per-question basis, and guidance associated with one task is not transferred across different questions, preventing cross-question information leakage and ensuring that reflective questioning remains grounded in the learner's current reasoning context. Unlike conventional similarity-based

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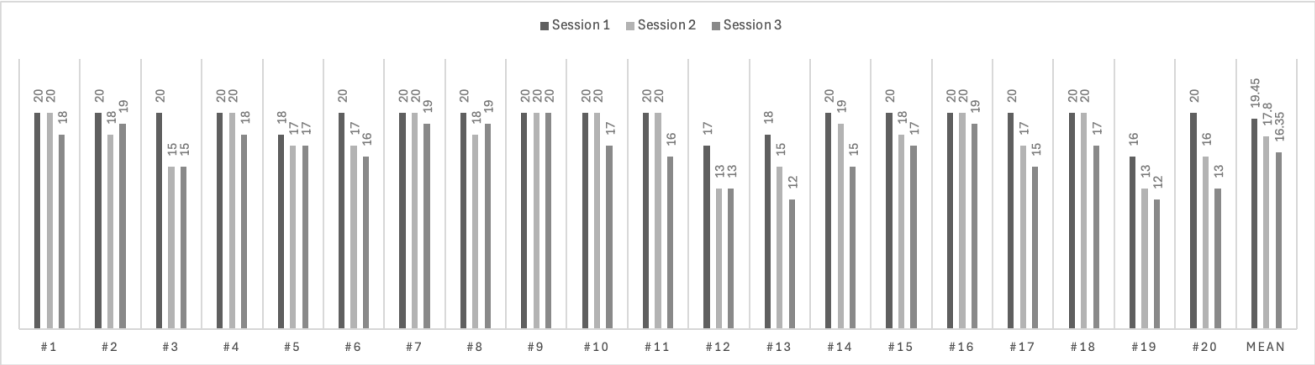


Fig. 2. Number of system-generated reflective questions required by each participant across learning sessions

Retrieval-Augmented Generation systems that retrieve information through semantic matching between user queries and document embeddings, retrieval in the proposed framework is instructionally controlled. Learning tasks and reasoning hints are directly selected from predefined PISA content units according to instructional design rather than similarity search, ensuring pedagogical consistency and preventing irrelevant information retrieval. By combining standardized PISA materials, randomized question selection, and semi-structured reasoning guidance, the RAG mechanism functions as a pedagogical control layer that constrains instructional content while enabling adaptive metacognitive facilitation through conversational interaction.

IV. Experiment and Result

A. Experimental Setting

A preliminary deployment study was conducted to examine the feasibility of the proposed metacognitive conversational framework in an authentic learning environment. Participants consisted of 20 undergraduate students undertaking cooperative education training at a technology-oriented organization. All participants interacted individually with the proposed system over three learning sessions, conducted at one-week intervals to allow reflection and potential adaptation of reasoning strategies across sessions. In each session, learners were presented with a PISA science assessment task selected from the repository described in Section 3.2.2. To ensure experimental control, all participants received the same initial question within each session, while subsequent interaction trajectories differed according to individual explanations and system-generated questioning. The task was presented as an open-ended explanatory problem requiring justification-based responses.

During interaction, the conversational agent generated reflective questions according to the CT-adaptive questioning framework summarized in Table 1. Each session continued until either (1) the system detected sufficient demonstration of structured computational thinking processes and terminated questioning automatically, or (2) a predefined upper limit of 20 reflective questions was reached. Participants responded freely to each system prompt without time restriction. To obtain an initial indicator of metacognitive development, the primary measurement collected was the number of reflective questions generated by the system per session. Since questioning is reduced when learners demonstrate stronger computational thinking characteristics, a lower number of generated questions indicates reduced reliance on metacognitive scaffolding during problem solving. This within-subject measure enables observation of changes in learners' interaction behavior across repeated system usage.

B. Results

Figure 2 presents the number of reflective questions generated by the conversational agent for each participant across the three learning sessions. Since questioning is reduced once learners demonstrate sufficient computational thinking characteristics, the number of system-generated questions represents the level of metacognitive scaffolding required during interaction.

During Session 1, nearly all participants required the maximum number of reflective prompts, indicating that substantial guidance was initially needed for learners to produce structured explanations. This suggests that participants were unfamiliar with the reflective reasoning processes encouraged by the system. In Session 2, questioning frequency decreased, with several learners completing the interaction before reaching the predefined prompt limit, indicating partial improvement in reasoning structure. The distribution of results also became more varied, reflecting differences in individual adaptation to metacognitive questioning. A further reduction occurred in Session 3, where most participants required fewer prompts before interaction termination. The mean number of generated questions decreased from 19.45 in Session 1 to 17.80 in Session 2 and 16.35 in Session 3, as shown in Figure 2. Because all sessions used identical initial PISA tasks, this reduction cannot be attributed to task difficulty. Instead, the results provide preliminary evidence that repeated engagement with CT-aligned reflective questioning may help learners develop more structured reasoning during explanation construction.

C. Interaction Case Analysis

To illustrate how the proposed framework operationalizes metacognitive scaffolding through computational thinking–aligned questioning, an interaction example derived from a PISA science task concerning animal cloning is presented. The learning content describes the cloning process used to produce Dolly the sheep and explains the involvement of three different sheep during the experiment. Learners were asked the following open-ended question: "Which sheep is Dolly identical to, and why?" Consistent with the interaction design described in Section 3, answer options were not provided, requiring learners to construct explanation-based responses.

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Case A: One learner stated: "Dolly is identical to all of the sheep because Dolly is a clone." Although the learner recognized the concept of cloning, the explanation lacked problem decomposition and causal reasoning regarding the roles of the three sheep in the experiment. The response treated cloning as a general concept without distinguishing the biological mechanisms involved. Instead of correcting the answer, the conversational agent generated a reflective prompt aligned with the decomposition dimension of computational thinking: "Each sheep played a different role in the experiment. Can you explain what role each sheep had in producing Dolly?" This question encouraged the learner to reconsider the structural relationships within the problem. Through subsequent revisions, the learner refined the explanation by distinguishing the nucleus donor, egg-cell donor, and surrogate mother roles.

Case B: Another learner responded: "Dolly is identical to sheep 1 because the nucleus came from sheep 1." This explanation correctly decomposed the experimental roles but showed limited abstraction regarding the underlying biological principle. The system therefore shifted questioning toward conceptual generalization rather than structural clarification. The conversational agent asked: "Why does transferring the nucleus make Dolly genetically identical even though the egg cell came from another sheep?" This prompt encouraged the learner to articulate the relationship between nuclear genetic material and organism characteristics, promoting abstraction beyond the specific scenario.

These examples demonstrate that questioning behavior varies according to reasoning characteristics rather than answer correctness. Learners exhibiting incomplete computational thinking receive structure-inducing prompts, whereas learners demonstrating partially correct reasoning are guided toward deeper conceptual abstraction. Interaction terminates once learners consistently exhibit coherent decomposition, abstraction, and logical reasoning, reflecting adaptive fading of metacognitive scaffolding. Together with the quantitative reduction in questioning frequency results previouusly shown, this example illustrates how the proposed conversational framework supports learners in progressively internalizing computational thinking practices through explanation-contingent reflective dialogue rather than answer provision.

V. Conclusion and Future Work

This paper presents a conversational learning framework that operationalizes Socratic-style inquiry using Large Language Models to support learners' metacognitive development. Instead of generating answers, the system engages learners in reflective dialogue guided by Computational Thinking (CT), prompting them to examine and refine their explanations during problem solving. By combining role-constrained LLM interaction with instructionally controlled retrieval of PISA science tasks, the framework demonstrates how conversational AI can implement inquiry-based learning in scalable environments. Deployment results show a gradual reduction in reflective questions across sessions, suggesting improved structured reasoning aligned with CT processes. Interaction analysis also indicates that adaptive questioning responds differently to incorrect, incomplete, and partially developed explanations, encouraging learners to reconstruct understanding through guided reflection rather than direct correction. Future work will extend the framework by incorporating additional thinking schemas such as systematic thinking, analyzing collected dialogues to refine Socratic questioning strategies, and investigating approaches to balance metacognitive questioning with information-providing agents while preserving learners' independent reasoning.

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